



CSE260: Digital Logic Design
Spring 2025
Quiz - 02
Duration: 30 Minutes

B

Name: <u>Solution</u>	ID:	Section:
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Instructions: Answer on the space provided.

Question 1 [CO2]: Convert the boolean function to its Canonical POS form. [7.5]

$$F(A,B,C,D) = A + B'C' + DB$$

$$= (A + B'C' + D)(A + B'C' + B)$$

$$= (A + D + B'C')(A + B + B'C')$$

$$= (A + D + B')(A + D + C')(A + B + B')(A + B + C')$$

$$= (A + B' + D + C')(A + C' + D + \overset{1}{B \cdot B})(A + B + C' + DD')$$

$$= (A + B' + C + D)(\overset{1}{A + B' + C' + D})(\overset{2}{A + B + C' + D})(\overset{1}{A + B' + C' + D})$$
$$\frac{(A + B + C' + D)(A + B + C' + D)}{2}$$

$$= (A + B' + C + D)(A + B' + C' + D)(A + B + C' + D)(A + B + C' + D')$$

$$= 0100, 0110, 0010, 0011$$

$$= \pi(4, 6, 2, 3)$$

$$= \pi(2, 3, 4, 6)$$

Question 2 [CO1]: Draw the following functions using NAND gates only. [7.5]

$$F(A,B,C) = B' + A \oplus C$$

$$= B' + A^2C + AC'$$

